

SAN FRANCISCO STATE UNIVERSITY

SCHOOL OF SOCIAL WORK

Interventions to Improve Mental Health Amongst Foster Youth

A Culminating Project

Presented in Partial Fulfillment of the Requirements

for the Degree of

MASTER OF SOCIAL WORK

By

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CERTIFICATION OF APPROVAL

I certify that I have read “Interventions to Improve Mental Health Amongst Foster Youth” and that in my opinion, this work meets the criteria for approving a thesis submitted in partial fulfillment of the requirements for the degree: Master of Social Work, Title IV-E at San Francisco State University.

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Abstract

The purpose of this study was to: (1) evaluate evidence-based treatments and practices used to treat PTSD experienced by foster youth, (2) to analyze if treatments and practices were accessible to the foster youth population, and (3) to identify limitations of each treatment. Foster youth experience many hardships while in foster care, leading many to need mental health services. While there are many therapies and interventions focused on improving mental health, what is not known is how successful they are in treating the foster youth population experiencing PTSD. This study assessed the effectiveness of five commonly used strength-based interventions that aid in improving foster youth mental health around PTSD. The questions used to guide this research were: (1) What are the successful evidence-informed interventions used to improve mental health amongst foster youth? (2) What are the factors of success that these interventions exhibit in improving mental health amongst foster youth? and (3) What are the gaps in current services of evidence-based treatments aimed at improving foster youths' mental health? This study conducted a comprehensive review that critically and systematically analyzed 45 peer-reviewed articles from the years 2000 through 2021. Evidence-based treatments that were included in this study were picked through the California Evidence-Based Clearinghouse which gave a rating for an evidence-based treatment's evidence and compatibility for working with child welfare involved youth. Articles that were peer-reviewed and about the topics of foster youth's mental health, PTSD, or the selected evidence-based treatments were selected and analyzed. A table was created that listed the authors' names, the year of the study, the study's samples and methods used, and a summary of its findings. Databases utilized for this study's article collection included: ProQuest Social Services Abstracts, Sociological Abstracts, Social Work Abstracts, PsychInfo, PsychExtra, and JSTOR. Implications for social work practice rose

from the findings displayed in Appendix 2. These findings served as a reference point that explains which treatment is most suitable for which symptoms of PTSD experienced. It also served as a reference point to anticipate barriers to services in addition to its factors of success to provide the practitioner with a thorough timeline of possible improvement. For social policy, findings encouraged the need and highlighted the lack of adequate training in interventions for all involved parties that play a role in reshaping the foster youth's life. For social research, findings showed a need to continue research on further methods of interventions and their longitudinal outcomes. These multidimensional implications served to gather existing information and corroborate them to further excel the use of evidence-based treatments in child welfare, serving the foster youth population, while also identifying the gaps or needs to excel them further.

Keywords: foster care, foster youth, evidence-based treatment, mental health, PTSD

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Chapter I. Introduction

Problem Statement

Youth in foster care experience high rates of trauma compared to their non-foster youth counterparts, contributing to a higher risk for mental health struggles and poor life outcomes. This is evident in how foster youth experience higher rates of adverse childhood experiences (ACEs) than non-foster youth (Hambrick, Oppenheim-Weller, N'zi, & Taussig, 2016). Foster youth face a higher risk of mental health and psychosocial challenges (Bederian-Gardner et al., 2018). Their risk for depression and anxiety is found to heighten the older they are (White et al., 2015). Research shows that children who experience plentiful moves while in long-term foster care are more likely to encounter behavioral and emotional problems, decreasing the likelihood of achieving any reunification or permanency (Font, 2014). Research also shows that 30% of foster youth have met the lifetime diagnostic criteria for Post Traumatic Stress Disorder (PTSD), a disorder that affects their ability to achieve self-sufficiency into adulthood (Gregory, 2021). This is a rate twice that of youth that never spent time in foster care (Gregory, 2021).

The instability and trauma that a majority of foster youth face are commonly associated with PTSD (Bederian-Gardner et al., 2018). While their life outcomes improve when social capital and support increase, foster youth are a population that is traditionally socially isolated with little to no familial support (Barnes-Proby, 2016). Studies have found that foster youth utilize mental health services 10–20 times more than youth in the general population, yet, the utilization of mental health services does not mean that youth were given adequate care or services (Villagrana, 2017), as they are also at a high risk of not receiving services that address their needs (Kerker & Dore, 2006). Reasons for the lack of services being provided may include the high financial costs, a health care system that values cost over care, frequent placement changes,

and even the lack of a person in their life who feels responsible to advocate for their well-being (Kerker & Dore, 2006). Society is also affected when the mental health needs of foster youth are not addressed, as if left untreated, these youth often find themselves homeless, in prisons, jails, institutions, or mental hospitals soon after emancipation (Kerker & Dore, 2006).

Foster youth are at higher risk of mental health issues due to many factors such as: experiencing child abuse and neglect, lack of relationship with parents, placement disruptions, lack of preparation for independent living, and multiple family problems (Shin, 2005). This population is faced with a lot of instability in their lives which is associated with PTSD (Bederian-Gardner et al., 2018). Youth who have mental health problems while in care are likely to continue to have those problems after they age out of the foster care system (Shin, 2005). It has also been found that 20% of foster alumni between the age of 20 to 33 have at least three psychiatric problems, compared to only 3% of the general population (Pecora, White, Jackson, L. & Wiggins, 2009). Foster youth alumni with a history of mental health problems are less likely to find employment and earn less than youth with no foster care history and mental health problems (Villegrana, 2017). While foster youth alumni have fairly high recovery rates from lifetime disorders, some mental health problems still have lower recovery rates, such as generalized anxiety disorder (39.6%), social phobia (26.6%), bulimia (25.8%), and PTSD (15.7%) (Pecora et al., 2009). These persisting mental disorders are also prone to comorbidity, as alumni with depression or PTSD also experience high rates of other illnesses, such as other emotional or behavioral disorders, or physical health conditions (Pecora et al., 2009). The higher rates of disorders among alumni compared to youth suggest that alumni of the foster care system may be more at risk for developing mental health problems than youth still in care, possibly due to unresolved issues surfacing in the difficult years after emancipation, when young adults may

not have the means or support to adequately handle them (Pecora et al., 2009). Assessing evidence-based practices aimed to improve foster youth mental health while in care is vital to stop this progression of poor mental health in foster youth and foster alumni. The prevalence rate of emotional health risks amongst foster youth defines the magnitude of this problem of poor mental health in foster youth. High numbers of displacement amongst foster youth are highly correlated with depression and feelings of hopelessness (Salazar, Keller, & Courtney, 2011). When these mental health needs continue to go unmet, children's emotional and behavioral difficulties worsen and lead to "burn out" of foster parents and caseworkers (Kerker & Dore, 2006). More demands are then placed on the child welfare system as it has to replace foster families and caseworkers more frequently (Kerker & Dore, 2006). The more severe a youth's behavior problems are, the more likely they are to experience higher rates of placement disruptions, creating a cycle of worsened mental health and more placement disruptions (Kerker & Dore, 2006).

Despite foster youth having high exposure to trauma and risk of PTSD, they often do not receive empirically validated interventions to mitigate the effects of their trauma (Mersky, Topitzes, & Blair, 2017). Services lacking to address their needs are contributed to financial costs and a fragmented health care system (Kerker & Dore, 2006). This contributes to fewer providers accepting Medicaid insurance or limiting the services they offer to those with that insurance (Kerker & Dore, 2006). Even when foster youth can receive mental health services, little is known about the mental health service delivery process (Baker, Schneiderman, & Licandro, 2017). There is a lack of knowledge of what interventions are being used in the delivery of mental health services to treat foster youth and how successful they are. This study searched for successful interventions to improve the mental health of foster youth before they

leave care and sustain them into adulthood. Strength-based interventions including Parent-Child Interaction Therapy, Cognitive Behavioral Therapy, Eye Movement Desensitization and Reprocessing Therapy, Child-Parent Psychotherapy, and Prolonged Exposure Therapy were examined and compared in a comprehensive literature analysis assessing their ability to reduce the symptoms of PTSD in foster youth. Evaluating evidence-based practices aimed at reducing symptoms of PTSD in foster youth can create a uniform practice model for service providers to use when working with this population.

Definitions of Terms

Child welfare involvement: Youth and families that are formally or informally involved in a continuum of services designed to protect children, strengthen families to care for their children, and promote permanency when children cannot remain with or return to their families (Children's Bureau Child Welfare Information Gateway, n.d.).

Evidence-based Treatments, Interventions, and Practices: Involves approaches to prevention or treatment that are validated by some form of documented scientific evidence. This includes findings established through controlled clinical studies, but other methods of establishing evidence are valid as well (Children's Bureau Child Welfare Information Gateway, n.d.).

Foster Care: A 24-hour substitute care for children placed away from their parents or guardians, and for whom the state agency has placement and care responsibility. This includes, but is not limited to, placements in foster family homes, foster homes of relatives, group homes, emergency shelters, residential facilities, child care institutions, and pre-adoptive homes (Children's Bureau Child Welfare Information Gateway, n.d.).

Foster youth: A child who has been placed in the state or county's legal custody because the child's custodial parents/guardians are unable to provide a safe family home due to abuse, neglect, or an inability to care for the child (Children's Bureau Child Welfare Information Gateway, n.d.).

Mental health: Mental health includes emotional, psychological, and social well-being and affects how one thinks, feels, and acts (Mentalhealth.gov, 2020)

Post-traumatic Stress Disorder: A mental health disorder that some people develop after experiencing or witnessing life-threatening or other traumatic events. PTSD is characterized by pervasive feelings of distress after the event. PTSD is diagnosed if the duration of symptoms is longer than a few months or if symptoms are significantly disrupting daily life (Children's Bureau Child Welfare Information Gateway, n.d.).

Objectives and Research Questions

The purpose of this study was to: (1) evaluate evidence-based treatments and practices used to treat PTSD experienced by foster youth, (2) to analyze if treatments and practices were accessible to the foster youth population, and (3) to identify limitations of each treatment. The purpose was also meant to evaluate which evidence-based treatments are most successful at decreasing the symptoms of PTSD and if they are accessible to the foster youth population, along with identifying what gaps still exist in limiting them from receiving evidence-based treatments. Overall, the goal was to learn what service providers are doing well and what can be improved in helping improve foster youth's mental health and enable them to successfully transition into functional adults.

This comprehensive literature review is guided by the questions:

- What were the successful evidence-informed interventions used to improve mental health in reducing the effects of PTSD amongst foster youth?
- What were the factors of success that these interventions exhibit in improving mental health and reducing the effects of PTSD amongst foster youth?
- What were the gaps in services of these interventions in improving foster youths' overall mental health and well-being?

Significance of Study

This study aimed to help evaluate evidence-based treatments for PTSD experienced by the foster youth population in hopes to reduce its long-lasting effects. It did this by evaluating therapies used to treat PTSD, assessing if they were appropriate for foster youth, and what their limitations were. The findings of this study gave guidance on how the child welfare system can better serve foster youth while in the system with improved mental health services and possibly creating access to more types of interventions. It could influence policy change, such as increasing Medicaid benefits for foster youth to have access to more mental health services and for extended periods. This could also influence changes in types of mental health services being offered by assessing which methods are most beneficial for this population and making those more accessible. There is a lack of research on what evidence-based treatments are used and accessed by foster youth. This study aimed to first assess what evidence-based treatments were appropriate for youth with child maltreatment history and with future research goals to assess if appropriate evidence-based treatments are regularly used within the field of child welfare. For social work practice and policy, it is crucial to understand how effective strength-based interventions are in decreasing poor mental health experienced by foster youth. It is important to build on existing research to seek alternative interventions that can support connections in

attachments in a child's life to work on strengthening relationships between parents, service workers, and foster children to achieve continuity and a sense of permanency (McLaughlin et al., 2015). Lastly, it is also important to furnish the child with a trauma-informed environment that will encourage them to feel comfortable and safe in reforming attachments and relationships that will protect their sense of stability and belonging in their placements while also bonding healthy emotional security in their life (O'Neill et al., 2012).

Chapter II. Theoretical Framework

Contemporary Trauma Theory

Contemporary trauma theory provided the basis for trauma-informed care in social work methods and practice by shifting how child welfare workers perceive and treat victims of trauma (Suleiman, 2008). The shift of focus is seen as child welfare workers move from asking, “what is wrong with the child?” to asking “what happened to the child?”. To elaborate further, child welfare workers fight to answer the question: What did the child witness or experience that had a powerful, traumatic effect on their mental and emotional health? By reframing the question to surround what happened rather than what is wrong, a shift in perspective occurs (Suleiman, 2008). The shift in perspective encourages child welfare workers to reframe how they view trauma-affected clients by seeing them as survivors who have been psychologically or physically injured and who need healing, rather than seeing them as poor functioning or as deficient (Mitchell, 2018). In relation to this study, the primary goal of contemporary trauma theory was to understand that poor mental and emotional health in current and former foster youth comes from unresolved psychological trauma.

When an individual witnesses, experiences, or is exposed to a traumatic event, long-lasting adverse effects hurt the individual’s mental and emotional health, as they must carry the weight of their trauma into their developing lives (Mitchell, 2018). In using contemporary trauma theory as a basis for trauma-informed care, guideline principles are formed to reduce the harsh effects of PTSD, and other related disorders such as anxiety, depression, or paranoia in current and former foster youth through the use of evidence-based intervention therapies so they can thrive in a life where trauma no longer rules them.

The guiding principles of contemporary trauma theory prioritize safety and choice while focusing on the three E's of trauma: events, experiences, and effects (Suleiman, 2008). In the first E, *events*, child welfare workers look at the list of events or incidents that may have had a traumatic impact on the individual. In the next E, *experiences*, child welfare workers investigate all experiences the individual has lived that were deemed physically or emotionally harmful, threatening, or overwhelming to the individuals' mental and emotional capacity. Here, it is taken into heavy consideration how the first E, events, were perceived in the individual's mind and how it is currently affecting them now. Lastly, in E for *effects*, child welfare workers investigate the severity of the lasting effects from these previously defined experiences on an individuals' mental and emotional developmental functioning. In using the three E's, child welfare workers can accurately assess a child's overall well-being, in hopes to reduce the harsh effects of PTSD in current and former foster youth so they can thrive in a life where trauma does not follow or slow them down (Ai, Jackson Foster, & Pecora, 2013). Contemporary trauma theory understands that often traumatic events, experiences, and effects, can lead to a maladaptive internalization of the incident altogether, disturbing an individual's biopsychosocial functioning, harming their mental and emotional health, and alter regions in the brain related to emotions, behaviors, and other areas of functioning.

In using contemporary trauma theory, child welfare workers can understand the level of impact a person's trauma has on their emotional and mental health and work sensitively to the child's early life experiences (Steenbakkers, Van Der Steen, Greitens, 2018). The benefits of using contemporary trauma theory concerning this study are seen in reducing the trauma impacts of child maltreatment and adverse childhood experiences. Contemporary trauma theory believes the events and experiences of a child's life develop a series of accumulative effects that interfere

with future life outcomes, therefore, it is necessary to work on reducing those effects through the use of evidence-based interventions and in trauma-informed care. According to Steenbakkers and Van Der Steen (2018) and their collection of data portraying how current and former foster youth view their child welfare experience, there is restored validation in the individual who experienced adverse childhood experiences when using a contemporary trauma theory lens approach to healing. Contemporary trauma theory encourages child welfare workers to utilize curative factors in mitigating the intensity of an individual's traumatic impact (Ai et al., 2013). Used as a protective factor in reducing the intensity of traumatic impact, contemporary trauma theory can also help in improving immune response and neurodevelopmental effects that hurt a child's cognitive, behavioral, and physical abilities (Forkey & Szilagyi, 2014). Such effects may include difficulty coping, managing cognitive processes, regulating behavior, and limited emotional intelligence due to struggles with PTSD. Contemporary trauma theory aids in addressing these areas of needs as it looks at the child's maltreatment and adverse childhood experiences as the reasoning foundation for the delay of overall well-being and development. Lastly, contemporary trauma theory partners in using mental health interventions to consider what the child has experienced in regards to maltreatment or trauma exposure, over what the child is exhibiting, to support vulnerable foster youth in their medical, mental health, and developmental needs (Hambrick et al., 2016). This theory guided this study by ensuring each evidence-based treatment selected contributed to the focus of treatment being, what happened to the child and what did the child experience, rather than what is wrong with the child when maladaptive behavior arose.

Social Learning Theory

Social learning theory proposes social behavior is learned by observing and imitating the modeled behavior, attitude, and emotional reactions of others. Developed by psychologist Albert Bandura (1977), social learning theory also suggests learning can occur through the observation of rewards and consequences, coined as vicarious reinforcement (Forenza, Bermea, & Rogers, 2018). This theory encourages the consideration of the environmental and cognitive factors that intersect to influence a child's learning and behavior in how they form and maintain healthy connections and relationships to people around them (Abbassi & Aslinia, 2010). In relation to this study, child welfare workers and current or prospective foster and biological parents can use social learning theory to identify who are the behavior models in the child's life and to what effect they are having an impact on the child and their mental and emotional health. In the mental health aspect, having appropriate models who model healthy human connections demonstrating proper responses and reactions to others is vital to enhance a child's psychological well-being (Hambrick et al., 2016).

According to Forenza and Bermea (2018), their research collected on foster youth's perceptions and abilities to form and maintain healthy relationships and connections with others, there is a high need for communication skill-building and comprehensive relationship education to improve their set perception (2018). Foster youth who have grown up in maltreatment and adverse childhoods do not learn how to appropriately form healthy connections with others, being their friends, dating partners, families, or teachers because they do not have the appropriate models in their homes in which adversity took place. There is great importance in incorporating social learning theory-based interventions to reeducate foster youth on how to manage and facilitate through positive partnerships during their prime developmental years, as well as in their

transitions to adulthood (Forenza et al., 2018). In the mental health aspect, foster youth who seek connections and relationships similar to the ones learned in their homes of adversity are more likely to encounter mental health complications as they are more likely to form unhealthy relationships (Hambrick et al., 2016). Therefore, foster youths need to relearn how to love and how to communicate with others to begin seeing an improvement in the mental health and human relationship axis. This relearning can be done in their new and current home placements where social learning theory based interventions are being incorporated.

Social learning theory based interventions are vital to connecting foster youth to new prospective homes and the caregivers leading those homes; they are also vital to encouraging foster youth in seeking healthy supports that contribute positively to their overall well-being (Abbassi & Aslinia, 2010). The resulting benefits from using social learning theory based interventions that increase overall well-being are seen in improvements with self-organization skills, increased engagement amongst the disengaged, and healthy collaboration amongst youth (Hambrick et al., 2016). In addition, social learning theory is best reinforced with child welfare workers and current or prospective foster parents as they are the prime adults in the foster child's life, therefore, are the prime models in which foster youth observe, study, and imitate (Van Holen, Vanschoonlandt, & Vanderfaeille 2016). Lastly, social learning theory based interventions can be used to increase the rapport each foster child has with the important adults in their homes and their service providers. With increased and strengthened rapport with positive models to follow, foster youth are more likely to thrive (Hambrick et al., 2016) This theory guided this study by encouraging the consideration of outside elements that may affect youth in need of evidence-based treatment. This theory ensured all evidence-based treatments selected weigh in external factors as barriers to treatment and how to overcome them for the best result.

Chapter III. Methods

Research Questions

The questions used to guide this research were:

- What were the successful evidence-informed interventions used to improve mental health that reduced the effects of PTSD amongst foster youth?
- What were the factors of success that these interventions exhibited in improved mental health and reduced the effects of PTSD amongst foster youth?
- What were the limitations in services of these interventions in improved foster youths' overall mental health and well-being?

Research Design

The present study conducted a comprehensive literature review on findings of evidence-based interventions that support foster youth's mental health in reducing the effects of PTSD in their lives. A comprehensive literature review provided a wide range of information that enabled this study to analyze existing therapies and assess for the most successful and effective methods and interventions used in treating PTSD and mental health issues. These interventions could be applied in therapeutic settings with either the foster child, the foster parent or current caregiver, child welfare workers, or other relevant service providers. A compare and contrast approach was used and presented in Appendix 2 to determine the usefulness and limitations of various evidence-based treatments. Appendix 2 also identified which mental health factors or behaviors the evidence-based treatment was best at reducing or improving (See end of Methods section). The strength of a comprehensive review was that it allowed a wide range of research to be able to be assessed and pulled information from, and didn't limit it to a single research article. The drawback to this design type was that it was limited by what data and

information were available in the already existing literature and was harder to gain information on the context of the data collected.

Search Strategy and Selection Criteria

Researchers utilized the San Francisco State University Library online website to gain access to article databases; Social Services Abstracts, Social Work Abstracts, Sociological Abstracts, PsycINFO, ProQuest, Ebscohost, Science Direct, Psych Articles, and ResearchGate. Keywords used to identify relevant articles included: foster care, foster youth, PTSD, mental health, and evidence-based treatment. For inclusion criteria, the following were reviewed: how to improve foster parent or caregiver relationships with foster children, as well as how to improve and ensure the quality of services by child welfare workers and other relevant service providers. Peer-reviewed journal articles that expanded the timeline of 2000 to 2021 were selected to be included in this research. Search strategy reviewed and compared related literature from various areas in and out of the country, drawing ideas and data from a diverse collection to best depict an accurate collection of what interventions are most effective relating to foster youth.

Evidence-based interventions and treatments included were determined by which have the highest evidence ranking for treating trauma and child welfare relevance scores according to the California Evidence-Based Clearinghouse for Child Welfare website.

Data Extraction & Procedures

This study conducted a comprehensive literature review that critically and systematically analyzed 45 peer-reviewed articles from the years 2000 through 2021. Evidence-based treatments that were included in this study were picked from the California Evidence-Based Clearinghouse which gave a rating for an evidence-based treatment's evidence and compatibility for working with child welfare involved youth. Peer-reviewed articles were thoroughly read and

reviewed on the topics of foster youth, trauma responses, effects of PTSD, evidence-based treatments, and overall mental health and well-being. Table 1 was created and consisted of the authors' names, the year of the study, the study's samples and methods used, and a summary of its findings. Appendix 2 was created and assessed each evidence-based treatment in terms of its evidence rating, relevancy to child welfare involved youth, gave a brief description of the therapy, age appropriateness, symptoms that it addressed, the estimated time frame for results, and any limitations it had. Databases utilized for this study's article collection included: Social Services Abstracts, Social Work Abstracts, Sociological Abstracts, PsycINFO, ProQuest, Ebscohost, Science Direct, Psych Articles, and ResearchGate.

Data Analysis

After articles were selected and read in-depth, the information was synthesized to examine interventions used to treat PTSD in foster youth. Within each article, the findings sections were paid close attention to in extracting data that was relevant to the research questions. These researchers organized each of the findings to answer the research questions into Table 1 to synthesize the information found. Each findings section was assessed and organized into categories relevant to each of the research questions. This study did not involve human subjects.

In the discussion of Appendix 2 titled, Evidence-based treatments for Child Welfare Involved Youth Experiencing PTSD, the purpose was to compress and create a visual of these researched trauma-informed evidence-based treatments. It also highlighted each one's beneficial value in treating or reducing PTSD and other mental health-related symptoms in foster youth by the child welfare worker or the foster parent caregiver in a comprehensive form. The titles of the investigated evidence-based interventions, practices, or treatments used in this study were listed

in the left column, divided into multiple rows. Along the top, heading each column, were the values that were identified and considered in regards to the listed titles of investigated evidence-based treatments or practices. These values included an evidence rating and child welfare reference, a brief description of the treatment, age-appropriateness or target population, the applicable symptoms treated, timeframes of treatment, goal beneficiaries or improvements of each treatment, and lastly, limitations in each treatment. In the use of Appendix 2, it was depicted which evidence-based interventions, practices, or treatment should be used to treat specific emotional and mental well-being amongst foster youth in following its brief description and timeline of use. The selected works could be utilized by any child welfare professional trained in identifying trauma and mental health needs amongst foster youth in the child welfare system due to reasons of child maltreatment or exposure to adverse childhood experiences.

Protection of human subjects

This study remained in alignment with Health Insurance Portability and Accountability Act protocols where applicable. Since this study utilized already collected data rather than collecting new data, no human subjects were a part of this study or at risk of harm and exposure. The Office of Research and Sponsored Programs at San Francisco State University approved this study. This study does not require further review because the project does not meet the federal definition of “human subjects” and was deemed “exempt” by the Institutional Review Board of the University (See Appendix 1).

Table 1. Selected Works

Literature Review Analysis				
Author (s)	Year	Sample	Type of Study	Summary of Findings
Ai et al.	2013	A critical review of the Midwest Study of Youth in Foster Care, The Casey Field Office Mental Health Study of Adolescents in Foster Care, The Northwest Foster Care Alumni Study, and The Midwest Study of Former Foster Youth.	Qualitative	Explores trauma consequences from PTSD in foster youth care and foster youth alumni.
Abbassi & Aslinia, S. D.	2010	Peer-reviewed articles related to family violence.	Literature Review	Children learn violent behaviors and continue to teach these behaviors to their offspring, perpetuating intergenerational trauma.
Agazzi et al.	2019	Recruited toddlers (n=8) and caregivers from local Part C and child welfare programs.	Mixed Methods	Assessed caregivers' ability to use appropriate social-communication skills with their children, to teach children to appropriately regulate and express strong emotions, to set healthy limits and expectations for children.
Baker et al.	2017	Children (n=156) receiving mental health services within the foster care agency	Quantitative	Predictors of length of treatment included being younger and being male.
Barnes-Proby	2016	Former foster youth in Santa Clara County	Mixed Methods	Results of this exploratory case study provide preliminary evidence of the utility of a networked approach to improving outcomes of adult foster youth.
Bederian-Gardner et al.	2018	Data for foster youth (n=146) that participated in the National Youth in Transition Database. Non-foster care youth (n=83) recruited from high schools that serve low socioeconomic status at-risk students.	Quantitative	Group comparisons revealed that instability was associated with posttraumatic stress disorder symptoms for foster but not non-foster youth.
Benbenishty et al.	2018	Data for this study were drawn from the 2011–2012 and 2012–2013 administration of the California Healthy Kids Survey in districts across California	Empirical Study; Quantitative Study	In-school experiences, such as in-school victimization, gang involvement, discrimination-based harassment, and cutting classes are responsible for many of the more negative academic outcomes experienced by foster youth.
Burns et al.	2004	NSCAW consists of two cohorts of children randomly selected to take part in this survey, along	Quantitative	Nearly half of the youths aged 2 to 14 years with completed child welfare investigations had clinically with significant emotional or behavioral problems. Only one-fourth of foster youths assessed to need specialty

		with their associated caregivers and child welfare workers. (n=3803) Children were aged 2 through 14 years		mental health care received any during the previous 12 months.
Cohen	2011	Analysis of 3 case studies of youth participating in TF-CBT.	Qualitative	Youth exposed to ongoing traumas and their non-offending parents have experienced significant improvement, indicating TF-CBT as an option for working with complex cases where violence risk is not readily or quickly eliminated.
Courtney & Society for Research in Child Development	2009	Examined the evolution of U.S. policy towards foster youth using the concept of "corporate parenting."	Qualitative	Remaining challenges of supporting foster youth still exist including: state reluctance to expand the parenting role; a poor knowledge base regarding the effectiveness of independent living services; lack of established and well-evaluated models of coordination between child welfare agencies and other public institutions in supporting foster youth; the complex nature of "permanency" for foster youth in transition; and exclusions of important populations of foster youth.
Craig & Sprang	2014	Children (n=216) ranging in age from 7 to 18 years who were receiving services at a university-based outpatient trauma treatment center.	Quantitative	Results suggest that female children start at higher reported total posttraumatic stress disorder rates than males, but both groups experience significant symptom reduction during the course of treatment. At post-treatment, girls are still reporting higher symptom levels than boys, suggesting that their clinical presentation at discharge may differ despite significant treatment gains.
Fernandez et al.	2004	236 school children who witnessed the Pirelli Tower crash and were treated through EMDR	Quantitative	A non-randomized trial of a group intervention of EMDR to 236 participants exhibiting PTSD symptoms after witnessing a tower crash. All but two children returned to normal functioning after EMDR. Distress Scale
Font	2014	Peer-reviewed case study on types of kin-related placement for foster youth in care.	Literature Review	Explored the positive effects on foster youth's overall well-being when removed from their homes and placed in kinship homes where they are familiar with the families and locations they are now in. Compared to foster youth statistics when placed out of home and out of related or familiar zones.
Forenza et al.	2018	Yout apart of a supportive housing program was in two focus groups (n = 5; n = 6), and individual interviews (n = 5)	Qualitative	Explored how teenaged foster youth managed relationships, facilitating positive partnerships in their transitions to adulthood.
Forkey & Szilagyi	2014	Analysis of case studies on foster youth in care and the effects of adverse experiences on mental and behavioral outcomes.	Literature Review	High impacts of complex trauma on foster youth's immune system and neurodevelopment along with cognitive, behavioral, and physical consequences complex trauma has on children who enter foster care with adverse childhood experiences.
Gilboa-Schechtm an et al.	2010	Adolescent w/ PTSD age 12-18	Qualitative	Treated youth no longer met diagnostic criteria for PTSD after treatment; this was also maintained at their 6 and 17-month follow-ups

Gonzalez	2014	Analysis of the effects of abuse and neglect on the socio-emotional and mental health outcomes of children.	Systemic Review	Draws attention to the most prevalent mental health conditions that affect foster youth and their families based on the type of maltreatment. Makes a push for further medical options and availability to be given to children and their families who suffered abuse.
Gregory	2021	Content analysis was accomplished using research and peer-reviewed studies and predefined criteria for inclusion.	Systematic Review	30% of foster youth met the lifetime diagnostic criteria for PTSD. The lifetime prevalence of PTSD for aged-out youth is twice that of their peers who never spent time in foster care. Over half of the prior foster care respondents were found to be using prescriptions for depressive disorders and experiencing psychological distress.
Hambrick et al.	2016	Analysis of peer-reviewed empirical studies of possibly efficacious interventions delivered to 0- to 12-year-old children in foster care.	Systematic Review	Most interventions have yet to be rigorously evaluated in community-based settings with children in foster care. Little research has been conducted on child and family engagement within these interventions, and there is a need for more research on moderators of intervention outcomes and subgroups that benefit most from these interventions. In addition, there is no consensus regarding how to adapt interventions to this population.
Hedenstrom	2014	Interviews of former foster youth (n=9) age 20-25 years old.	Qualitative	1) 'Facing a transition' which had three sub-themes of 'abrupt transition,' 'smooth but uneven transition,' and 'smooth transition;' 2) 'Depending on adults for help' which had two sub-themes of 'receiving support' and 'asking for support;' 3) 'Making a difference for others, and 4) 'Moving forward with my life.' These four themes highlighted the participants' experiences as they aged out of foster care.
Ippen	2011	Preschool-aged children (n=75) and their mothers referred to treatment following the child's exposure to domestic violence.	Mixed Methods	Children in the 4+ and <4 risk groups, those who received CPP showed significant improvements in PTSD and depression symptoms. Analyses of 6-month follow-up data suggest improvements were maintained for the high-risk group.
Johnson & Pryce	2013	Utilized existing outcome data from a client outcome database (n=262) to evaluate whether participation in Therapeutic Mentoring as part of the service delivery system decreased trauma symptoms for youth in foster care.	Quantitative	Analysis suggests that mentored youth who remain in the program longer (up to 18 months) and receive TM improved significantly over non-mentored youth in terms of demonstrating a reduction in the expression of stress symptoms associated with trauma.
Kerker & Dore	2006	Review of the literature on the mental health of foster children.	Literature Review	Foster youth are at an increased risk of not receiving services necessary to address their needs due to financial costs, and a fragmented health care system, and because they often lack a person in their life who feels responsible and accountable for their well-being. Untreated mental disorders among children in the child welfare system impact not only children's functioning and well-being but also society.

Kerns et al.	2014	Surveyed and conducted 16 focus groups. Participants had a history of child welfare involvement (n=127) from 27 of the 48 state field offices.	Mixed Methods	Identified challenges of caseworkers having limited training in identifying mental health needs, uncertainty about how to apply screening results to case planning, and concerns about the roles, relationships, and communication with mental health providers, variability in the extent of available evidence-based services. Mental health providers indicate multiple challenges of difficulty communicating and coordinating with child welfare caseworkers and families and lack of training on the child welfare system and the unique needs of children and youths in foster care.
Lesperance	2018	A longitudinal study of aged out youths (n=57) during an 8-year period.	Quantitative and Qualitative	As foster alumni get older, they have decreased in time spent homeless and deviant behaviors and increases in income from employment over time. They had no change in risky sexual behaviors, psychological distress, or victimization was found, indicating sources of persistent difficulties among these youth.
Leve et al.	2009	Presented evidence from 4 randomized clinical trials that included components of the Multidimensional Treatment Foster Care program.	Qualitative	Showed a need for mental health services to be expanded to include multidimensional treatments.
Lotty et al.	2019	Standardized assessment measures were used on the intervention group (n = 49), and control group (n = 30).	Quasi-experimental	Evaluated effectiveness of Foster Connections Program, trauma-informed care based intervention highlighted to support Foster Care Parents in their capacity to provide care for children who have high ACES
McLaughlin et al.	2015	Research findings guiding social policy discovering importance of social worker continuity.	Literature Review	Explores the disconnect between child welfare workers, social workers, and social justice towards children and foster youth in services. Explores opportunities to mend and strengthen the disconnect for improved quality of life and services.
Mersky et al.	2017	Utilized supporting evidence from a community-university partnership in Wisconsin.	Qualitative	Foster youth rarely receive empirically validated interventions to mitigate the effects of trauma such as PCIT.
Mitchell	2018	Longitudinal research study examining the outcomes and experiences of youth transitioning out of foster care (n=200).	Quantitative	Explores the effects of Non-Death Loss in Foster Youth such as ambiguous loss, symbolic loss, and disenfranchised loss. Also explores its impact on FY psychological and emotional well-being.
O'Neill	2012	Performs review on results of stability with fewer or more placement disruptions for youth in care in terms of development.	Literature Review	Locates aspects of development affected by placement disruptions and stability. Stability explored in aspects of home, school, friend and family relationships, continuity of child welfare or social worker.
Pecora	2001	Analysis of Casey National Alumni Study, Northwest	Quantitative	Factors that have been linked with educational success in foster youth include pursuing permanency to help youth

		Alumni Study, and Salazar Study of College Completers.		find enduring mentors, maximizing placement and school stability, conducting a strengths-based assessment, aggressively pursuing educational supports, and treating mental health problems that may act as barriers to classroom success.
Pecora et al.	2009	Performs data review on the emotional, behavioral and substance abuse disorders of children in foster care and former recipients of foster care.	Systematic Review	The higher rates of disorders among alumni compared to youth suggest that alumni of foster care may be more at risk for mental health problems than youth still in care. This may be because unresolved issues surface in the difficult years after emancipation, when young adults may not have the means or support to address them properly. Emphasis on careful screening and assessment methods for youth entering and remaining in foster care is made.
Purvis	2015	Randomized sample (n=96), pre-post design with a control group.	Qualitative	Evaluated effectiveness of trauma-informed, attachment-based intervention taught to current and ex foster children's parents in reducing behavioral problems and trauma symptoms in their children.
Rubin et al.	2001	39 participants in the randomized control trial	Qualitative	The study evaluated the effectiveness of adding EMDR. Clients were randomly assigned to receive EMDR plus the center's routine treatment package, or a control group that received only the center's routine treatment package. Analyses found no differences between the overall groups. Analyses on clients with elevated baseline scores found moderate effect sizes that approached but fell short of, statistical significance. Limitations include the small sample size, the diverse nature of diagnoses in the sample, and the lack of a trauma-specific outcome measure.
Salazar et al.	2011	Analysis of depression and low feelings of hopelessness amongst foster youth experiencing displacement.	Literature Review	In an analysis of depression and low feelings correlated to displacement, views on understanding the importance and need for social support for the affected child following a list of adverse experiences.
Settersson & Ray	2010	A systematic review of articles and reports assessing the time it takes for youth to leave home and gain independence.	Systematic Review	Families tend to support their children for an extended period. The continued need to rely on families for financial assistance exacerbates the plight of young people from a variety of vulnerable backgrounds, such as foster youth.
Shin	2005	A random sample of foster youth (n=113) in a large Midwest state.	Quantitative	Foster youth who had experienced abuse were about 23xs more likely to receive mental health services.
Steenbakkers et al.	2017	Sample of thirteen youth and young adults who had to have at least one foster family placement that lasted two years, and had to be an older adolescent or young adult.	Qualitative	High impacts of trauma experience for youth, the impact of quality of Child Welfare services, social and emotional problems FY face, and search for stability amongst foster care parents.
Suleiman	2008	Paper on Trauma Theory.	Qualitative	Explains the use of CTT in providing interventions and services to Foster Youth in hopes to reduce the effects of PTSD in children of care.

Timmer et al.	2006	Non-relative foster parents and their foster children (n=75) compared with non-abusive biological parent (n=98)	Quantitative	Showed decreases in child behavior problems & caregiver distress from pre- to post-treatment for both foster & biological parent-child dyads, revealing no differences in the effectiveness of this treatment for foster parents compared with non-abusive biological parents.
Van Holen et al.	2017	A randomized controlled trial with foster families (n=200).	Qualitative	Evaluates effectiveness of trauma-based intervention-based Social Learning Theory for parents who care for foster children exhibiting behaviors.
Villagrana	2017	17-year-old and older youth (n=934) who were in the foster care system.	Quantitative	Racial/ethnic disparities in mental health service use upon exit from the foster care system were found, with Latino youth using the least amount of services after foster care exit. Racial/ ethnic service disparities in the type of services used were also found.
Weiner et al.	2009	Utilized the Child and Adolescent Needs and Strengths on a diverse sample of youth ages 3–18 at six child welfare involved agencies.	Quantitative	The compared evidence-based treatments effectively reduced symptoms and improved functioning among minority youth. Implementation issues, including challenges and culturally competent accommodations, are discussed.
White et al.	2015	Foster care alumni (n=65) between the ages 23 and 24 of Michigan.	Quantitative	Foster care alumni had similar or worse life outcomes than their peers and significantly worse outcomes in the areas of mental health, substance dependence, and education.

Chapter IV. Results

A. Trauma-Focused Evidence-Based Treatments

The following therapies were found to be effective trauma-focused evidence-based treatments as determined by the California Evidence-Based Clearinghouse for Child Welfare. This national website for mental health acts as a reference guide for practitioners to utilize. Evidence-based treatments that were rated lower were selected, as the lower the score, the greater level of research support there was for that evidence-based treatment. The description sections depicts the evidence-based treatment, the age range it could be used with, and the length of time required for each therapy. The effectiveness section of each evidence-based treatment describes what factors of success the treatment has in improving mental health and reducing the effects of PTSD amongst foster youth. The limitation section of each treatment assessed the disadvantages of the therapy. After depicting each evidence-based treatment, this study discussed common barriers to foster youth receiving mental health and evidence-based treatment services were examined to understand where the gaps in services are within the current system.

Parent-Child Interaction Therapy

Description: Parent–Child Interaction Therapy (PCIT) is an evidence-based treatment for children with behavior problems who have experienced abuse and caregivers experiencing distress (Timmer, Urquiza, & Zebell, 2006). It was developed by Dr. Sheila Eyberg in the 1970s, for children ages 2–7 (Mersky et al., 2017) but has also been found beneficial to reduce externalizing symptoms in children 2-12 years of age and their families where physical abuse and family violence were prevalent (Craig & Sprang, 2014). Its focus on working with children who experienced trauma makes it suitable for youth with PTSD. PCIT helps aid caregivers in managing difficult behaviors of their foster children by building their relationship with the youth

and teaching the caregivers behavior management skills (Timmer et al., 2006). PCIT is based around social learning theory and its principles of observing and imitating modeled behavior, attitude, and emotional reactions of others (Timmer et al., 2006). PCIT incorporates both the caregiver and child in sessions, with the first phase of sessions focusing on enhancing the parent-child relationship and the second phase focusing on improving child compliance (Timmer et al., 2006). Both the first and second phases of treatment have an hour of didactic training, followed by therapist coaching in structured role-play situations (Timmer et al., 2006). The first phase is based around child-directed interaction where parents are taught to give their children praises, reflect their speech, imitate their behavior, describe their behavior, and enjoy playing with their children (Mersky et al., 2017). The second phase is based around parent-directed interaction (Timmer et al., 2006) which teaches caregivers disciplinary practices and behavior management skills that help promote their child's emotional and behavioral regulation (Mersky et al., 2017). Through parent-directed interaction, caregivers focus their attention on their children's positive behaviors while learning to give clear and direct commands (Mersky et al., 2017). Once caregivers can give effective commands, they learn to provide praise for compliance and strategies for dealing with non-compliance (Mersky et al., 2017). The increased bond between caregiver and child promotes symptom stabilization, using the caregiver-child relationship and behavior modification as the vehicle for change (Craig & Sprang, 2014).

Effectiveness: PCIT targets young children because they are sensitive to corrective experience and their behaviors are significantly influenced by caregivers during this period (Mersky et al., 2017). PCIT has shown to be an effective evidence-based treatment with its positive effects being maintained for up to six years post-treatment (Timmer et al., 2006). Its effects can be seen in children in school settings and their untreated siblings (Timmer et al.,

2006). PCIT has been used as a clinical treatment for children with emotional and behavioral dysregulation that display externalizing and acting out behaviors like aggression, oppositional behavior, defiance, and non-compliance (Timmer et al., 2006) (Mersky et al., 2017). Timer et al. (2006), completed a two-phase study that included biological and non-relative foster parents with a history of maltreatment to assess the compatibility for PCIT use with families involved in the child welfare system. The sample in the first phase was 385 participants that were made up of children, foster parents, and biological parents and the second phase consisted of 173 participants (Timmer et al., 2006). Caregivers completed the assessment-based protocol and participated in PCIT, and were not considered done with treatment until they were able to meet mastery criteria for the treatment (Timmer et al., 2006). Overall, PCIT was found to have beneficial effects for foster parents who completed treatment with their foster children, showing a biological relationship is not required for this treatment (Timmer et al., 2006). With studies successfully replicating PCIT findings with children a part of the child welfare system, it is clear that it can be further beneficial to this population. It offers large effect sizes and relatively modest cost per client compared to alternative evidence-based treatments (Mersky et al., 2017).

Limitations: While PCIT is shown to help reduce symptoms of PTSD by strengthening the relationship between child and caregiver, it is limited to children of a young age. This evidence-based treatment is not inclusive of children with delays and disabilities, excluding many children (Agazzi, Ferron, Ferron, Shaffer-Hudkins, & Salloum, 2019). The length of this treatment is also high, as 12-14 weekly sessions can be a lot for foster parents to attend. This evidence-based treatment also lacks an evidence rating from the California Evidence-Based Clearinghouse as a trauma-informed practice but has shown effectiveness in aiding children and their families experiencing symptoms of PTSD.

Trauma-Focused Cognitive Behavioral Therapy

Description: Trauma-Focused Cognitive Behavioral Therapy (TF-CBT), is an evidence-based treatment for children between the ages of 3 and 18 to treat anxiety and PTSD, cognitive and affective dysregulation, and behavioral and interpersonal problems (Mersky et al., 2017). While it is effective for youth up to 18 years old, it is ideal for school-aged children who have experienced a discrete traumatic experience or series of experiences (Weiner, Schneider, & Lyons, 2009). This treatment is based on the creation of a “trauma narrative” that allows the client to build a narrative around the traumatic event (Weiner et al., 2009). TF-CBT focuses on parental education to normalize the youth’s and caregiver’s reactions to severe stress, provide information about psychological and physiological reactions to stress, and educate families about the benefits and need for early treatment (Weiner et al., 2009). The treatment attempts to reduce physiologic manifestations of stress and PTSD by explaining the body’s response to stress and teaching relaxation techniques such as focused breathing, mindfulness, meditation, and physical activity (Weiner et al., 2009). It also teaches youth skills useful for coping with ongoing trauma, including awareness of and increased ease with coping with ongoing trauma reminders and recognizing traumatic responses (Cohen, 2011). TF-CBT also utilizes desensitization, cognitive strategies, and safety planning (Weiner et al., 2009). This treatment requires the children to interact with the therapist and their caregivers in an information-sharing, affiliative, and collaborative manner, a skill that may be easier for girls than boys (Craig & Sprang, 2014). There is a significant body of empirical literature that points to the efficacy of TF-CBT for reducing trauma symptoms in children suffering from PTSD as a result of sexual abuse (Craig & Sprang, 2014). TF-CBT more directly targets the cognitive processing and schema disruptions identified

in emotional processing theory, suggesting a direct relationship between treatment goals and symptom outcomes (Craig & Sprang, 2014)

Effectiveness: Within a study done by Weiner et al. (2014) a sample of 35 of their 213 participants were enrolled in TF-CBT. The Child and Adolescent Needs and Strengths tool was utilized to assess the children's initial status and used for follow-up assessments after beginning treatment (Weiner et al., 2009). TF-CBT had shown significant reductions in “Traumatic Stress Symptoms”, “Behavioral/Emotional Needs” and significant increases in “Strengths” (Weiner et al., 2009). It was effective with a diverse ethnic and racial population along with a wide range of ages (Weiner et al., 2009). TF-CBT can also be used to treat past and ongoing trauma, making it an option for working with ongoing cases where violence risk is not already or will be quickly eliminated (Cohen, 2011). It has also been shown effective for sexual violence (Craig & Sprang, 2014).

Limitations: TF-CBT requires 12-18 weekly sessions and can be a lot for caregivers of foster youth to attend with the youth. Since it is shown effective for a large age range, the training for this TF-CBT comes into question how well providers are trained for different age and developmental stages and how the training considers this. It also may be more effective for girls, due to their higher ease of information sharing.

Child-Parent Psychotherapy

Description: Child-Parent Psychotherapy (CPP) has been linked to improved outcomes for infants and toddlers, ages 0-5, in high-risk families by successfully addressing attachment and regulatory problems (Mersky et al., 2017). The child is seen with their primary caregiver, and the pair are treated together (California Evidence-Based Clearinghouse, 2015). CPP examines how the trauma and the caregivers' relational history affect the caregiver's relationship

with the child and the child's development (California Evidence-Based Clearinghouse, 2015). The main goal is to strengthen the caregiver-child relationship as a vehicle for restoring and protecting the child's mental health by supporting improved communication and understanding (California Evidence-Based Clearinghouse, 2015). Treatment also focuses on contextual factors such as culture, socioeconomic status, and immigration-related stressors that may affect the relationship (California Evidence-Based Clearinghouse, 2015). Throughout treatment, the caregiver and child are guided to create a joint narrative of the psychological traumatic event and identify and address traumatic triggers that lead to or cause dysregulated behaviors and affect (California Evidence-Based Clearinghouse, 2015).

Effectiveness: Within a study done by Weiner et al. (2014) a sample of 65 of their 213 participants were enrolled in CPP. The Child and Adolescent Needs and Strengths tool was utilized to assess the children's initial wellness status and used for follow-up assessments after beginning treatment (Weiner et al., 2009). CPP shows to be appropriate for diverse populations who experienced trauma, as, within the study, biracial youth experienced significant improvements in "Traumatic Stress Symptoms," "Strengths, Behavioral/Emotional Needs," and "Risk Behaviors", while Hispanic youth experienced significant improvement in "Traumatic Stress Symptoms," "Life Domain Functioning," and "Behavioral Emotional Needs" (Weiner et al., 2009). White youth improved significantly in "Life Domain Functioning" (Weiner et al., 2009). CPP has been shown to promote safe behavior, improve affect regulation, and build the relationship between caregiver and child (California Evidence-Based Clearinghouse, 2015). It's a reinforcement of behaviors that help caregivers and children cope with their trauma and gain a new perspective on their feelings and experiences (California Evidence-Based Clearinghouse, 2015).

Limitations: CPP is a less regimented treatment implemented with very young children, so while certain components are included, it is not feasible to document specific elements of each interaction, lowering its reliability (Weiner et al., 2009). The requirement for participants to be below the age of five also limits this evidence-based treatment from being utilized by a large proportion of foster youth and caregivers. It also requires a long treatment period of 52 weekly sessions. This year's long commitment may be too much for many caregivers of foster youth to make, and waiting a year to see benefits may not help youth in time before a placement change occurs.

Prolonged Exposure Therapy for Adolescents

Description: Prolonged Exposure Therapy (PE), is a cognitive-based behavioral treatment aimed to reduce the overall effects of PTSD and other mental health-related dysregulations such as depression, anger, guilt, and general anxiety (California Evidence-Based Clearinghouse, 2015). In PE, the treated subjects are encouraged to repeatedly approach situations or activities they are prone to avoid because they are reminders of their experienced trauma coming from previous child maltreatment or exposure to adverse childhood experiences. Treated subjects can approach situations or activities related to facing their traumas through either in vivo exposure, in actuality, or imaginal exposure, through retelling it until a different effect or impact is achieved. The goal of using these two exposure types is to create a gateway for subjects to safely process their traumatic memories. Another benefit comes as treated subjects learn coping skills to be less reactive when confronted with their trauma reminders or triggers in everyday life (Leve, Fisher, & Chamberlain, 2009).

Effectiveness: PE treatment is recommended to occur once or twice a week for approximately 8 to 15 sessions, if not 2 to 4 months. Each PE session should be 60 to 90 minutes

in length to produce its full effect in reducing trauma impacts on their subjects, in this case, the subjects being foster youth. Homework is assigned to the treated subject related to the therapeutic intervention practiced in each weeks' session; homework is designed accordingly to align with either the in vivo exposure or the imaginal exposure technique. With this format of treatment, effectiveness has been measured in assessing the well-being of the treated subject before and after beginning treatment. In one study, it was shown that the treated subjects demonstrated a great decrease in PTSD-related symptoms and an increase in global functioning after following the said treatment plan. Out of 68.4% of the treated subjects, composed of adolescent youth, 36.8% of those no longer met diagnostic criteria for PTSD after treatment; this was also maintained at their 6 and 17-month follow-ups (Gilboa-Schechtman et al., 2010).

Limitations: PE was treated to a group of adolescents, however, it is unclear how much of this group of adolescents was composed of foster youth who have endured child maltreatment or have been exposed to adverse childhood experiences.

Eye Movement Desensitization and Reprocessing

Description: Eye Movement Desensitization and Reprocessing (EMDR), is a type of psychotherapy designed to provide treatment to children and adolescents in need of trauma processing following detrimental exposure who are now exhibiting signs of PTSD and other mental health-related disorders (California Evidence-Based Clearinghouse, 2015). EMDR comes with standardized procedures in which the treated subject is guided through emotionally disturbing material in brief sequential doses that include the clients' beliefs, emotions, and bodily sensations associated with their trauma and the event that caused it while focusing on an external stimulus. In further description of EMDR, there are a total of 8 phases in which the guided practitioner directs the treated subject through the bilateral back and forward eye movements to

external stimuli, such as hand tapping, as the subject recalls events of their trauma. The goal of EMDR is to target past or current events or situations that trigger their current livelihoods in causing a disturbance, reduce their subjective distress and bodily response towards the traumatic event identified, and to lastly, promote rewiring so that the trauma memory is changed to a source of resilience (Rubin et al., 2001). The 8 phases of EMDR therapy begin with an assessment of the subjects' readiness for memory processing along with the development of a treatment plan. In the second phase, the working practitioner enlists coping, calming, and self-soothing skills the treated subject has acquired to use as a stabilizing mechanism when needed in the following three to seven phases. In the next three to seven phases, a target memory of a traumatic event is identified, as well as an image representing the most haunted part of the traumatic event. Once done, the act of reprocessing the memory begins by having the treated subject focus on themselves, their body, and any thoughts or sensations that come through as they begin to re-experience each set of bilateral stimulation. After each session, the subject is reminded to close by engaging in previously identified coping, calming, or self-soothing skills from the second phase (California Evidence-Based Clearinghouse, 2015).

Effectiveness: In one study of 236 participants, there was noted improvement in how the treated subjects bodily and mentally responded to traumatic events brought up in their 8 phases of EMDR. In-depth, this study was conducted with a sample of 236 school children treated in EMDR after witnessing a fatal tower crash produced similar positive results. The 236 children treated were each exhibiting symptoms of PTSD throughout 30 days following the crash, yet showed significant four-month improvement after the implementation of EMDR therapy techniques. After four months, every single child participant, with an exception of two children, demonstrated a return to normal functioning which was scaled with the Subjective Units of

Distress Scale (Fernandez, Gallinari, & Lorenzetti, 2004). A Subjective Units of Distress Scale was used to measure the intensity of distress one feels when exposed to certain stimuli that trigger PTSD or anxiety. The distress scale also takes into consideration one's bodily and mental responses to said triggers. In following this treatment plan, its effectiveness was shown in reducing PTSD, anxiety, fear, and other mental health-related disorders amongst a collection of treated subjects.

Limitations: EMDR was treated to a group of adolescents, however, it is unclear how much of this group of adolescents was composed of foster youth who have endured child maltreatment or have been exposed to adverse childhood experiences. Other limitations are found when considering the possible length of treatment as there is no accurate number of how long subjects should need EMDR therapy; it depends solely on the individual and the severities of their experience with multiple traumas, if not one. However, notable signs of effectiveness in successfully reducing the impact of trauma upon subjects are usually seen in just a few weeks of 3-12 sessions (California Evidence-Based Clearinghouse, 2015).

B. Barriers in Foster Youth Receiving Services

Micro Level

Demographic Trends: There is a high need for mental health services for youth involved with child welfare, but despite this high need, these youth are at an increased risk of not receiving services (Kerker & Dore, 2006). Those that do receive services also are not guaranteed to receive services that are adequate for their needs, for instance, of youth that exhibit a need for clinical mental health services, only a quarter of them receive clinical care (Burns et al., 2004) (Villagrana, 2017). There are also disparities found in which youth are referred to and utilize services while in care. Older, male, and White youth were more likely to receive mental health

services, along with youth who experienced sexual abuse (Burns et al., 2004) (Villagrana, 2017). Youth that experienced physical abuse were only referred when they displayed externalizing behaviors and children who experienced neglect were the least likely to utilize services (Villagrana, 2017). These studies show that there are trends in which youth utilize mental health services through a lack of why is unclear. This could point toward bias within who is referred by service providers, a lack of screening processes, or other unknown factors.

Interpersonal and Cultural Issues: Studies have found disparities in mental health usage in foster youth by racial/ethnic groups (Villagrana, 2017). African American and Latino youth in foster care are less likely to utilize mental health services than their White counterparts and Latino foster care youth are the least likely (Villagrana, 2017). Disparities such as these may be due to cultural differences in viewing mental health or lack of culturally diverse services and providers. A decreased usage of mental health services by 60% is seen after foster youth leave care (Villagrana, 2017). Despite this decrease in services, foster alumni have higher rates of disorders compared to youth still in care (Pecora et al., 2009). Pecora suggests that this may be because they face issues during the difficult years after emancipation, when these young adults may not have the means or support to address them properly (Pecora et al., 2009). It is also becoming harder for young people in general to attain independence and is seen in how caregivers are supporting their children for extended periods longer than historically seen (Settersson & Ray, 2010). This need for help can only be more evident for vulnerable youth, such as foster youth, who may not have caregivers or family members to rely on despite the societal struggle to achieve independence (Settersson & Ray, 2010). Being independent without means for a fallback could be the push for disorders to surface in their emancipated years, making a successful transition into adulthood even harder. Caseworkers perceived that previous negative

experiences with therapy, placement changes, wariness around new adults, or feelings of stigma for receiving mental health services were common barriers for foster youths (Kerns et al., 2014).

Mezzo Level

Accessibility of Services to Foster Youth: While individual foster youth face various challenges in getting the mental health services that they need, they also face issues even when they can access these services (Kerker & Dore, 2006). Foster youth tend to have high rates of placement disruption which often results in inconsistency in medical care provision (Kerker & Dore, 2006). Foster youth with higher emotional and behavioral difficulties often change placements at higher rates and with that, change providers as well, despite their higher need for consistency in services (Kerker & Dore, 2006). This also leads to providers not knowing their foster youth patients well enough to effectively advocate for their needs or coordinate their care (Kerker & Dore, 2006). It's also been found that few public child welfare agencies systematically screen children entering care for mental health problems, allowing for many to fall through the cracks (Kerker & Dore, 2006). The availability of services varies widely across states, with urban areas reporting the greatest variety of treatments and resources and smaller communities severely lacking (Kerns et al., 2014).

Availability of Service Providers: There is a lack of mental health professionals to provide services to the foster youth population, which is seen due to issues with Medicaid health insurance (Kerker & Dore, 2006). Since Medicaid often offers low reimbursement rates and puts caps on the number of services that can be provided to youth, few providers are willing to treat foster children whose needs may require more time than what they will be compensated for (Kerker & Dore, 2006). Kerker suggests increasing Medicaid reimbursement rates for treating foster children to enable more mental health providers to treat this population (Kerker & Dore,

2006). Providers have also reported wanting more training on the child welfare system and the unique needs of children and youths in foster care (Kerns et al., 2014). Studies have also shown that increased inter-organizational relationships between child welfare and mental health offices were associated with an increased likelihood that children and youths received mental health services (Kerns et al., 2014). Each additional inter-organizational relationship increases that likelihood by 4% (Kerns et al., 2014). Caseworkers also report needing help identifying agencies that provide evidence-based treatments that are relevant for child welfare involved youth and how to match agency services or evidence-based treatments with the needs of a specific youth (Kerns et al., 2014). Many caseworkers believe that their communities lack sufficient high-quality services and practitioners who understood the needs of the foster care population (Kerns et al., 2014).

Macro Level

Systemic Gaps and Failures: Many reasons including high financial costs, and a cost over care health care system, contribute to foster youths' mental health needs being unmet (Kerker & Dore, 2006). There has been an emphasis on the need for comprehensive, continuous care by individuals who are familiar with the special behavioral health needs of the foster care population, but there is little evidence to date that these recommendations have been implemented (Kerker & Dore, 2006). It's also been found that children that enter the child welfare system seldom receive empirically validated interventions to mitigate the effects of trauma. (Mersky et al., 2017). Their school district is also a required source of evidence-based treatment that often fights to not provide services to these youth (Kerker & Dore, 2006). A child's home school district is required to pay for evidence-based treatments, yet school administrators often do not want to pay for foster youth who are no longer attending their school

(Kerker & Dore, 2006). The youths' new school district may not want to pay for a child who may soon move on to yet another school (Kerker & Dore, 2006). Contributing to the lack of continuous care may also be due to foster youth often lacking a person in their life who is consistently responsible and aware of their mental health needs to properly advocate for the youth (Kerker & Dore, 2006). This could be seen as the fault of child welfare agencies who are unable to secure permanent placements for youth and results in youth experiencing multiple placements disruptions, contributing to worsened mental health in the youth. Many foster care agencies provide their youth with mental health services based on the assumption that they have an emotional or behavioral disorder, yet without the correct assessments to determine if they do (Pecora et al., 2009). When these youth's mental disorders go without treatment, they impact their functioning and well-being along with society (Kerker & Dore, 2006). When youth age out of foster care with untreated mental health disorders, they are at high risk of many poor life outcomes (Kerker & Dore, 2006).

Medicaid Coverage: Most children in foster care qualify for Medicaid health insurance, a federally funded but state-run health insurance program (Kerker & Dore, 2006). Medicaid provides modest provider reimbursements, with bureaucratic inefficiencies, and has led to few providers accepting this insurance due to inconvenience and lack of pay (Kerker & Dore, 2006). The few providers who do accept Medicaid are often restricted in the number of sessions that they will be reimbursed for to treat foster children's complex needs (Kerker & Dore, 2006). In addition, for Medicaid to pay for more extensive treatments, it often requires youth to have a Diagnostic and Statistical Manual of Mental Disorders diagnosis or measured evidence of a serious emotional or behavior disorder (Kerker & Dore, 2006). This restricts services to only the youth with the highest need or for providers to be more apt to diagnose a youth for them to get

the services they need (Kerker & Dore, 2006). It also allows for youth with less severe mental health needs to go untreated until they worsen to the point of eligibility (Kerker & Dore, 2006). Prevention and early intervention services are also limited in availability, so even those who are eligible encounter difficulties enrolling in such programs (Kerker & Dore, 2006). This leaves youth who enter care at an early age to have their needs missed and left to progress into larger problems before being addressed. Kerker suggests that federal insurance programs should broaden the conditions they cover to include children with less severe symptoms or with the kinds of early life experiences that are predictive of later emotional and behavioral difficulties regardless of current psychosocial functioning (Kerker & Dore, 2006). Also, the limits on the care that children receive by the number of appointments or services provided should be removed. (Kerker & Dore, 2006)

Chapter V: Discussion

The purpose of this study was to identify successful evidence-informed interventions and evaluate their usefulness in improving mental health in reducing the effects of PTSD and other symptoms amongst foster youth. The evidence-based treatment interventions found under this purpose include PCIT, TF-CBT, CPP, PE, and lastly, EMDR. In the evaluation of their usage or implementation, it was found that the listed therapies were successful as they focused on treating the problem that was defined by what the child experienced and lived through by considering all past and present external factors that played a role in shaping a child's life experience. Leading into the next aspect of this study's purpose, also investigated were the factors of success that these interventions exhibit in improving mental health and reducing the effects of PTSD amongst foster youth. Factors of success were found in resulting effectiveness of each therapeutic intervention using a child welfare reference scale used to determine effectiveness by examining the goals of the program, target population for which the program was developed, and any research that had been conducted for the program to determine its relevance to children and families served by the child welfare system. TF-CBT and CPP both resulted in a high child welfare reference score. A high child welfare reference score means the program was designed or is commonly used, to meet the needs of children, youth, young adults, and/or families receiving child welfare services. PCIT, PE, and EMDR all resulted in a medium child welfare reference score. A medium child welfare reference score means the program was designed, or is commonly used, to serve children, youth, young adults, and/or families who are similar to child welfare populations and likely include current and former child welfare services recipients. Factors of success by each intervention also varied based on which mental health symptom they most have improvement record on. PCIT is great to address non-compliant behavior, aggression,

disruptive behavior, and dysfunctional attachments to parents. TF-CBT is the best treatment for feelings of shame or distorted beliefs about oneself. PE has factors of success in treating PTSD symptoms, while EMDR has factors of success in treating PTSD symptoms and anxiety or fears. Lastly, CPP factors success in treating exposure to trauma through the internalization or externalization of PTSD symptoms. In contrast to investigating the factors of success, the purpose of this study continued to investigate gaps in services of these interventions in improving foster youths' overall mental health and well-being. The gaps of services were noted as the limitations of each investigated evidence-based treatment, creating barriers in who could benefit from its therapeutic participation. In PCIT, limitations were found in a small age range, limited language access, and lack of treatment findings specified to PTSD in child welfare. In TF-CBT, limitations were found in their wide age range and lack of adequate training for official implementation. PE and EMDR therapy both lack in-depth application to child welfare youth. In addition, EMDR pertains to gaps in services as the timeframe of treatment includes a wide variance in which results are seen. Lastly, CPP finds gaps in services in limited language access, as well as limited age range, and long treatment time. These findings provided a conclusion to the study's hypotheses and research questions in its reports of treatment results based on their levels and gaps of effectiveness. Through the conduction of this study, in which evidence-based treatments were identified, researched for implementation effectiveness, reviewed for factors of success, and reviewed for gaps of services, implications for future social work practice, policy, and research were released for further consideration.

Chapter VI. Implications for Social Work Practice, Policy, and Research

Compressing this research and findings, implications for social work practice, policy, and research could be taken to encourage the use of trauma-informed and evidence-based interventions to improve a foster child's experience in the child welfare system. In doing so, the goal is to reduce the effects of PTSD and other mental health-related symptoms following their experience with child maltreatment or exposure to adverse childhood experiences. These multidimensional implications each serve to address and improve continuity across all thresholds that play roles in positively influencing foster youth experience in the child welfare system.

Social Work Practice

For social work practice, the findings of this research aimed to guide child welfare workers, service providers, and current or prospective foster parents to use the most effective trauma-informed and evidence-based interventions, such as PCIT, TF-CBT, PE, EMDR, and CPP in practice to protect foster youth from further mental health consequences and their overall mental and emotional well-being. For child welfare workers and service providers in practice, an identified need for communication and relationship building in foster youth clientele could be highlighted before requesting such trauma-informed interventions and practices be implemented closely in the child's environment. Once the need to reduce PTSD and other mental health-related symptoms in foster youth are identified, child welfare workers and service providers are encouraged to not only implement the interventions themselves when working with the child, but also teach them to foster parents, educators, and other caregivers involved in the child's life to ensure there is continuity across services (Ai et al., 2013). For current or prospective foster parents and their practice in parenting at-risk foster youth, such evidence-based treatments and practices when implemented can improve foster youths'

comprehensive relationship education. With a need identified to improve comprehensive relationship education, foster parents will play a vital role in reshaping how the foster child connects with their role as a child, and others' role as caregivers (Gonzalez, 2014). There is a goal in improving relationship education between foster youth and their foster parents to ensure foster youth are retaught how to hold appropriate connections with others after coming from homes where these connections were not appropriately modeled (Forenza et al., 2018). This goal could be achieved in the reformation of how foster parents as caregivers use social work practice in their ways of parenting. In improving the way child welfare workers, service providers, and current or prospective foster parents conduct their practice related to their foster clients or children, the hope for foster children's mental and emotional well-being is long term (Gonzalez, 2014). In the long term, the goal is for foster youth to achieve an ability to thrive in a life where their past traumatic and other mental health-related impacts no longer deter their life skills. The overall goal for social work practice is to encourage all providers involved in child welfare to maintain and increase the steps towards a client-focused practice where mental and emotional needs are considered first to find their ability to thrive.

Social Policy

For social work policy, the findings of this research highlight the importance to require child welfare workers, service providers, and current or prospective foster parents to be adequately trained in trauma-informed care, as well as evidence-based treatments to reduce a foster child's mental health-related traumatic impact. In requiring so, effective and evolved parent training focused on implementing such interventions, that serve to help and strengthen bonds between the foster parent to the foster child, can improve the effectiveness of their involvement in reparenting children who have lived or been exposed to adverse childhood

experiences (Purvis et al. 2015). In addition, policies in parent training are also encouraged to require therapeutic mentoring for foster youth to reduce their trauma and mental health-related symptoms (Johnson & Pryce, 2013). Lastly, social work policies that require parent training must know that to honor the act of fostering youth in the child welfare system, it also means to foster connections in their home, school, and other close environments (Lotty, Dunn-Galvin, & Bantry-White, 2020). These environments are where there are opportunities to influence foster youths' abilities to thrive in working to reduce their traumatic and mental health-related symptoms (Lotty, Dunn-Galvin, & Bantry-White, 2020). Also for social work policy, there is hope in requesting an increase in funding for programs in the expansion of mental health services that aim to expose foster youth to evidence-based interventions. An increase in funding of mental health services cannot only improve the quality of services being provided but also maximize educational and overall life success in foster youth by supporting their ability to thrive over any adverse childhood experience they have been exposed to (Pecora, 2001). Lastly, there is great importance in ensuring that the increase of funding for mental health services that treat foster youth and their well-being are completely multidimensional in setting foster youth up for success following past child maltreatment or adverse childhood experiences (Leve et al., 2009).

Social Work Research

For social work research, these findings evaluated the importance of continued investigation of other useful interventions that were not addressed in this study and the potential positive impacts they may have. Future research should also focus on improving mental health amongst foster youth through effective trauma-informed and evidence-based interventions to reduce overall traumatic and mental health impact. Due to the complex nature of mental health and how every youth responds to treatment differently, researching a larger arsenal of techniques

or interventions to treat PTSD and other mental health disorders in foster youth is needed (Agazzi, Adams, Ferron E., Ferron J., Shaffer-Hudkins, & Salloum., 2019). The focus of further research aimed to reduce PTSD and other mental health-related symptoms in foster youth must focus on the impact that child maltreatment and adverse childhood experiences have on the psychological development of youth (Johnson & Pryce, 2013). In continuing further research, it is also important to consider possible longitudinal outcomes that come from implementing trauma-informed and evidence-based interventions (Lesperance, T. M., 2018). In searching for longitudinal outcomes, it is important to look at what will improve foster youths' success in their well-being and their ability to thrive in their current situations that will follow them as they transition out of care in becoming independent adults. In addition, further opportunities of multidimensional research should be done to investigate which interventions are being used and which are effective in addressing mental health services for transitional aged foster youth or have already exited care (Leve et al., 2009).

Chapter VII: Limitations of Study

The study had several limitations that created barriers in the execution of these findings. The evidence-based treatments that were selected for further investigation in this study share the same understanding in defining the problem as what the child has experienced, not the child itself, as well as considering all external and environmental influential elements that contribute to the problem. Evidence-based treatments outside of that scope were not chosen or evaluated for their possible effectiveness. The researched trauma-informed and evidence-based interventions chosen in this study were assessed for effectiveness in treating only a selective type of mental health and trauma-related symptoms. To define selective, there may be other emotional or behavioral symptoms related to mental health the youth may display that were not thoroughly covered within this study. Limitations were also found as some trauma-informed and evidence-based interventions assessed were similar to one another and may produce the same results. Other methodological and conceptual weaknesses predicted were found as most foster youth experience comorbidity of mental health dysregulations, therefore, it remains unknown if the researched interventions are treating the symptom listed, or if treating others. For example, as PCIT could be used in treating and reducing the effects of PTSD in foster youth, it could also be used to better relations and attachments, therefore, it is unable to tell which of the two symptoms it is improving. Lastly, this study was only focused specifically on the child welfare population, composed of foster youth in the child welfare system who have lived or been exposed to child maltreatment or adverse childhood experiences. Thus, the findings of this study may not apply to the youth of other populations who may be exhibiting the same mental and emotional dysregulations explored in this study.

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Appendix 1

IRB Approval Letter

Dear Graduates Studies,

Based on the information that Kimberlee Duncan-Rodriguez provided for project titled "*Interventions to Improve Mental Health amongst Foster Youth*" ORSP-HAP has determined that no further IRB review is required because the project does not meet the federal definition of "Not Human Subjects Research (If this option is checked, the study does not require IRB review, is not eligible for exemption)" per 45 CFR 46 102 (d): Research means a systematic investigation, including research development, testing and evaluation, designed to develop or contribute to generalizable knowledge.

Regards,

Human and Animal Protections, ORSP

Appendix 2

Evidence-Based Treatments for Child Welfare Involved Youth Experiencing PTSD

Evidence-Based Practice ^A	Evidence Rating ^B/ Child Welfare Reference ^C	Brief Description of EBT	Age	Symptoms Addressed	Time Frame	Limitations
Parent-Child Interaction Therapy	NA / Medium	It teaches parents traditional play-therapy skills to use as social reinforcers of positive child behavior and traditional behavior management skills to decrease negative child behavior. Parents are taught and practice these skills with their children in a playroom while coached by a therapist. The coaching provides parents with immediate feedback on their use of the new parenting skills, which enables them to apply the skills correctly and master them rapidly. ^D	2-7	Noncompliance, aggression, rule-breaking, disruptive behavior, dysfunctional attachment with a parent, internalizing symptoms ^D	12-14 weekly Sessions	-Limited training to only English and Spanish -Small Age Range -Lacks in-depth application to child welfare youth -Use for PTSD is new and lacks evidence rating for PTSD application
Trauma-Focused Cognitive Behavioral Therapy	1 / High	TF-CBT is a conjoint child and parent psychotherapy model for children who are experiencing significant emotional and behavioral difficulties related to traumatic life events. It is a component-based hybrid treatment model that incorporates trauma-sensitive interventions with cognitive-behavioral, family, and humanistic principles. ^D	3-18	Feelings of shame, distorted beliefs about self and others, acting out behavior problems, and PTSD and related symptoms ^D	12-18 Weeks	-Wide age range, the practicality of adequate training is unclear

Prolonged Exposure Therapy for Adolescents	1 / Medium	PE is a therapeutic treatment where clients are encouraged to repeatedly approach situations or activities they are avoiding because they remind them of their trauma (in vivo exposure) as well as to revisit the traumatic memory several times through retelling it (imaginal exposure). ^d	6-18+	PTSD and related symptoms ^d	8-15 sessions or 2-4 months	-Lacks in-depth application to child welfare youth
Eye Movement Desensitization and Reprocessing	1 / Medium	EMDR therapy is an 8-phase psychotherapy treatment that was designed to alleviate the symptoms of trauma. During the EMDR trauma processing phases, guided by standardized procedures, the client attends to emotionally disturbing material in brief sequential doses that include the client's beliefs, emotions, and body sensations associated with the traumatic event while simultaneously focusing on an external stimulus. ^d	2-18+	PTSD, anxiety, fears, and behavioral problems ^d	The length of treatment is impossible to predict and is dependent upon the severity of the trauma. Gains are often apparent within a few weeks ranging from 3-12 sessions.	-Lacks in-depth application to child welfare youth -Lacks clear timeframe of treatment with wide variance in when results are seen
Child-Parent Psychotherapy	2/ High	CPP examines how the trauma and the caregivers' relational history affect the caregiver-child relationship and the child's developmental trajectory. A central goal is to support and strengthen the caregiver-child relationship as a vehicle for restoring and protecting the child's mental health. Treatment also focuses on contextual factors that may affect the caregiver-child relationship (e.g., culture and socioeconomic and immigration-related stressors). Targets of the intervention include caregivers' and children's maladaptive representations of themselves and each other and interactions and behaviors that interfere with the child's mental health. Over the course of treatment, the caregiver and child are guided to create a joint narrative of the psychological traumatic event and identify and address traumatic triggers that generate dysregulated behaviors and affect. ^d	0-5	Exposure to trauma, internalizing and externalizing symptoms, and/or symptoms of PTSD ^d	52 weeks	-Lacks training in any language other than English, limits diversity -Applicable to a small proportion of youth -Long treatment time frame -Less regimented treatment

^A **Evidence-Based Practice:** A combination of the following three factors: (1) best research evidence, (2) best clinical experience, and (3) consistent with patient values

^B **Evidence Rating:** The purpose of the CEBC Scientific Rating scale is to evaluate each practice based on the available research evidence. A lower score indicates a greater level of research support.

^C **Child Welfare Reference:** The Child Welfare System Relevance Level is determined by examining the goals of the program, the target population for which the program was developed, and any research that has been conducted for the program to determine its relevance to children and families served by the child welfare system.

High: The program was designed, or is commonly used, to meet the needs of children, youth, young adults, and/or families receiving child welfare services.

Medium: The program was designed, or is commonly used, to serve children, youth, young adults, and/or families who are similar to child welfare populations and likely include current and former child welfare services recipients.

Low: The program was designed, or is commonly used, to serve children, youth, young adults, and/or families with little or no apparent similarity to the child welfare services population.

^D *California Evidence-Based Clearinghouse. (2015). Program Registry. Retrieved March 15, 2021, from <https://www.cebc4cw.org/program/>*